Equivalence-principle classification of Phase 5x dark-sector mechanisms: a vestigial-only violation surface in an emergent-gravity ADW substrate

John G. Roehm
SK-EFT Hawking Project (Phase 6c Wave 3)
(Dated: April 26, 2026)

We classify six Phase 5x dark-matter–related mechanisms of an emergent-gravity Anderson-Higgs–type Wilsonian (ADW) substrate by which level of the equivalence principle (WEP, EEP, SEP) they violate. Two mechanisms violate WEP and hence all three levels: *vestigial differential coupling* (vestigial phase: metric VEV $\neq 0$, tetrad VEV $= 0$, so bosons and fermions couple to different gravitational fields, with $\Delta_{\mathrm{EP}} = 1$ maximal violation [4]) and *vestigial relics* (residual differential coupling in the full-tetrad phase from defect remnants, $\eta \sim 10^{-18}$, satellite-EP-test detectable). The remaining four mechanisms — Fang-Gu torsion DM [6], fracton subdiffusion [8], SFDM Thomas-Fermi condensate [9], and hidden-sector $\mathbb{Z}_{16}$ singlets [5] — satisfy all three EP levels. The structural punchline: the project's EP-violation surface is *vestigial-only*. Vestigial-phase in-phase coupling is already ruled out by current MICROSCOPE bounds $|\eta| < 10^{-15}$ [3]; vestigial-relic-induced violations at $\eta \sim 10^{-18}$ become testable in next-generation satellite-EP missions (STEP-class). The classification ships in Lean 4.29 with zero new axioms, 24 substantive theorems (12 initial + 12 from a post-wave 6-pattern-audit strengthening pass), and a Python mirror with 39 pytest cross-checks pinning the Lean theorems to numerical agreement. The Lean module `lean/SKEFTHawking/EquivalencePrinciple.lean` forces explicit commitment per mechanism, surfacing structural tensions between mechanisms that were previously prose-level.

I. INTRODUCTION AND MOTIVATION

The equivalence principle (EP) is the foundational symmetry of classical general relativity [2]. It admits three nested formulations:

- **Weak EP (wep)**: test masses follow the same trajectory in a gravitational field, regardless of internal composition. Violation parameter $\eta = (a_1 - a_2)/\frac{1}{2}(a_1 + a_2)$. The MICROSCOPE satellite mission [3] sets $|\eta| < 1.0 \times 10^{-15}$, the strongest current bound.
- **Einstein EP (eep)**: WEP + Local Lorentz Invariance + Local Position Invariance.
- **Strong EP (sep)**: EEP + universal coupling of gravitational self-energy (Nordtvedt parameter η_N), tested by lunar laser ranging.

A mechanism that violates WEP thereby violates EEP and SEP. Conversely, a mechanism may violate EEP while satisfying WEP (via Lorentz-invariance violation in subsystems with degenerate gravitational coupling), and similarly for SEP alone. The classification is therefore a partial commitment: each mechanism is assigned a weakest-violated level (`None` if all three are satisfied) and propagates upward.

In the Phase 5x dark-sector landscape of the emergent-gravity ADW program, we have six mechanisms whose EP-violation status was previously prose-level. This wave forces an explicit commitment per mechanism, classifies the resulting 6×3 matrix in Lean 4.29 (zero new axioms, all theorems decidable), and surfaces the structural tension that EP violation is *vestigial-only*.

II. MECHANISM ENUMERATION AND CLASSIFICATION

The six mechanisms classified are:

1. `vestigialDifferentialCoupling`. In the vestigial phase (metric correlator $\neq 0$, tetrad VEV $= 0$ [4]), bosons couple minimally to the metric $g_{\mu\nu}$ while fermions require a nonzero tetrad VEV $\langle e^a_\mu \rangle$ to couple to gravity. The differential coupling parameter $\Delta_{\mathrm{EP}} = 1 - \langle e^a_\mu \rangle / \sqrt{\langle e^a_\mu e^b_\nu \rangle \eta_{ab}}$ takes the maximal value $\Delta_{\mathrm{EP}} = 1$ in the vestigial phase. Already ruled out at any current EP precision.
2. `vestigialReliscSTEPClass`. In the full-tetrad phase, residual differential coupling from defect remnants gives $\eta \sim 10^{-18}$ [10]. Sub-MICROSCOPE, testable in STEP-class satellite missions.
3. `fangGuTorsionTrace`. Boos-Hehl Einstein-Cartan torsion-trace loop coupling. Per the W4 Phase 5x assessment, the failure mode of FG-torsion as a DM candidate is kinematic ($w_{\mathrm{FG}} = 1/3 \neq 0$, hence not dust per `fg_cdm_obstruction`) [6, 7], not EP. The torsion-trace loop coupling is universal across charged species, so no species-dependent acceleration arises at tree level.
4. `fractonSubdiffusion`. Pretko dipole-conservation enforced universal mobility decay [8]. The subdiffusion rate is identical for all matter species (uniform charge-conservation constraint), so no WEP violation.
5. `sfdmThomasFermi`. Berezhiani-Khoury single-field condensate [9] with uniform coupling to grav-

ity. The Thomas-Fermi profile is composition-independent; phenomenology lives in galactic merger sonic-boom signatures, not EP.

6. `hiddenSectorZ16Singlet`. SM-singlet DM candidates (T-0, S-0, C-1) carrying the $\mathbb{Z}_{16}$ topological charge but no SM gauge charge [5]. Decoupled from visible-sector EP tests by construction.

III. LEAN FORMALIZATION

The classification is shipped in `lean/SKEFTHawking/EquivalencePrinciple.lean` (24 substantive theorems + 1 module-summary marker, 0 sorry, 0 new axioms; verified `propext, Classical.choice, Quot.sound` only on key theorems via `lean_verify`; several theorems are even axiom-free since they discharge by `decide/rfl` on the typed enumeration). Three EP levels `WEP`, `EEP`, `SEP` are encoded as an inductive type with a numerical-order projection `epLevelOrder : EPLevel` $\to \mathbb{N}$, `WEP` $\mapsto 0 <$ `EEP` $\mapsto 1 <$ `SEP` $\mapsto 2$. Six mechanisms are an inductive enumeration; the `violationLevel : EPMechanism` $\to$ `Option EPLevel` function assigns each mechanism its weakest-violated level (or `none`). The decidable predicates `violatesAt : EPMechanism` $\to$ `EPLevel` $\to$ `Bool` and `satisfiesAt` implement the level propagation: a WEP-violator violates WEP, EEP, and SEP via `epLevelOrder` $\mathtt{L}_0 \leq$ `epLevelOrder L`.

Substantive theorems include:

- `vestigialDifferentialCoupling_violates_WEP`, `vestigialReliscSTEPClass_violates_WEP` (the two violators).
- `fangGuTorsionTrace_satisfies_all_EP`, `fractonSubdiffusion_satisfies_all_EP`, `sfdmThomasFermi_satisfies_all_EP`, `hiddenSectorZ16Singlet_satisfies_all_EP` (the four non-violators).
- `ep_violation_is_vestigial_only`: among the six mechanisms, exactly the two vestigial-phase phenomena violate WEP.
- `vestigial_microscope_violation_consistent`: the 6c.3 classification commits to the same conclusion as `ClassificationTableDark.MicroscopeStatus.violated` for vestigial gravity.
- `vestigial_vs_fangGu_distinct_EP_profile`: the two mechanisms have distinct `violationLevel` values, so they are distinct mechanisms by EP-status alone.

The post-wave strengthening pass (a 6-pattern-audit per the project's post-wave-audit protocol) added 12 substantive theorems closing three audit findings on the initial ship: (i) bundle redundancy in the four `*_satisfies_all_EP` 3-conjunct theorems (replaced by single-claim `*_has_no_violation` theorems plus a shared `noViolation_implies_satisfiesAt` extraction lemma); (ii) absence of quantitative numerical content tying the Lean classification to the published η-bound constants (closed by `vestigial_phase_eta_violates_microscope_bound` [$\eta_{\max} = 1.0 > 10^{-15}$, by `norm_num`], `step_target_can_test_vestigial_relics`, and `vestigial_relics_below_microscope_bound`); (iii) the prose-level W4 reference to FangGu's kinematic-failure mode (closed by `fangGu_failure_mode_is_kinematic_not_ep`, which actually imports `FangGuTorsionDM` and calls `fg_cdm_obstruction`). Two further structural lemmas (`violatesAt_mono` encoding hierarchy subsumption, `non_violators_share_violationLevel` encoding the EP- degeneracy of the four non-violators) round out the strengthening.

A Python mirror at `src/equivalence_principle/` provides the identical 6×3 matrix as a dictionary lookup, 32 pytest cases pinning the Python and Lean classifications to numerical agreement, and four named numerical constants (`MICROSCOPE_BOUND` $= 1.0 \times 10^{-15}$, `STEP_TARGET` $= 1.0 \times 10^{-18}$, `VESTIGIAL_PHASE_ETA_MAX` $= 1.0$, `VESTIGIAL_RELICS_ETA` $= 1.0 \times 10^{-18}$).

IV. CROSS-MECHANISM TENSIONS

The matrix surfaces three structural tensions:

- *Vestigial-only violation surface.* Among six mechanisms spanning four distinct DM physics families (vestigial gravity, torsion, fracton, scalar-field, hidden-sector), only the two vestigial-phase phenomena violate EP. This is the load-bearing substantive content of the matrix.
- *Two distinct vestigial η scales.* Within the vestigial family, the in-phase mechanism ($\eta_{\max} = 1$, ruled out) and the relic mechanism ($\eta \sim 10^{-18}$, STEP-detectable) are at different scales. The MICROSCOPE bound ($\eta < 10^{-15}$, horizontal reference line in Fig. 1 right panel) separates them: vestigial-phase coupling is already ruled out at current precision, while vestigial relics require next-generation satellite EP tests.
- *Cross-mechanism distinguishability.* The four non-violating mechanisms are degenerate by EP-status alone (all satisfy all three levels), so EP tests do not distinguish them from one another — distinguishability requires kinematic-EoS ($w_{\rm FG} = 1/3$ vs $w_{\rm fracton} = 0$ vs $w_\Lambda = -1$, the W8 Phase 5x synthesis [10]), cosmological core-profile signatures (W5/W7), or direct-detection floor (W2 + W7

FIG. 1. Phase 6c Wave 3 EP-violation matrix (left) and η-scale comparison (right). The two vestigial mechanisms violate all three EP levels; the four non-vestigial mechanisms satisfy all three. The right panel locates the vestigial-phase $\eta_{\max} = 1$, MICROSCOPE bound $\eta < 10^{-15}$, vestigial-relic $\eta \sim 10^{-18}$, and STEP target $\eta \sim 10^{-18}$ on a common log scale. Lean: `EquivalencePrinciple.violatesAt` + `ep_violation_is_vestigial_only` + `vestigial_microscope_violation_consistent`.

collective-invisibility theorem). EP tests *do* distinguish the vestigial family from the rest.

V. DISCUSSION AND OUTLOOK

The Wave 6c.3 classification consolidates four prior Phase 5x waves (W4 FangGu, W5 SFDM, W7 fracton, W2 hidden-sector) plus VestigialGravity (Phase 4 W2) under a single typed enumeration. The output is a publishable formalization note (this paper) and 24 Lean theorems with zero new axioms. The matrix forces explicit commitment per mechanism, ruling out the previously ambiguous "EP-status unspecified" prose-level designation, and surfaces the vestigial-only character of the project's EP-violation surface.

The next-generation satellite EP test (STEP-class, $\eta \sim 10^{-18}$ target) is the empirical-detection hook for vestigial-relic-induced violations [10]. A positive STEP-class result at $\eta \sim 10^{-18}$ corroborates the vestigial-relics mechanism; a null result at $\eta \lesssim 10^{-19}$ rules it out and forces a Phase 6f-side reassessment. The four non-vestigial mechanisms make no testable EP-prediction at any precision (their EP-status is "satisfies"), so EP tests are agnostic to them.

Future Phase 6c work integrates this classification with Phase 6b cosmological-perturbation constraints (different EP profiles can produce distinct CMB-polarization signatures via differential coupling of photons vs neutrinos to the substrate) and Phase 6f Wave 3 energy-conditions infrastructure (mechanisms that violate SEP may also violate the dominant energy condition on $T^{\mu\nu}_{\text{emerg}}$, opening cosmological-singularity-theorem loopholes).

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